

A wild-dynamics operator on every infinite-dimensional Banach space

fa_banach_001, lane 7

2026-06-20

Abstract

Augé proved that every infinite-dimensional separable Banach space admits an operator $R = I + K$, with K compact and even nuclear, for which

$$A_R = \{x : \|R^t x\| \rightarrow \infty\}$$

has non-empty interior but is not dense; equivalently, A_R and a recurrent set B_R form a partition of the space. Remark 3.8 of the source paper conjectures that the same construction should be possible in arbitrary Banach spaces. This packet proves that conjectural extension. The only extra ingredient is to replace the separable bounded biorthogonal system by a normalized basic sequence and to choose the periods so fast that the periodic part returns to the identity in operator norm.

1 Source question

The source is Jean-Matthieu Augé, *Linear operators with wild dynamics*, arXiv:1204.2044. Theorem 1.1 proves the desired wild-dynamics construction for separable Banach spaces. The open-ended extension appears in Remark 3.8 on page 8:

from the definition of R that for any $\epsilon > 0$, one can choose the sequence (m_k) to have $\|R - I\| \leq \epsilon$. This proves that I is in the norm closure of $\mathcal{A}_X$.

REMARK 3.8. Although Corollary 1.2 does not apply to ℓ^∞ , we can see that our example works in ℓ^∞ . We thus conjecture that operators with the properties stated in Theorem 1.1 can be constructed in arbitrary Banach spaces.

3.2. Real case. We now show the version of Theorem 1.1 for real spaces.

In the notation of the source paper, $\mathcal{A}_X$ denotes the class of operators for which A_R and

$$B_R = \{x : \varliminf_{t \rightarrow \infty} \|R^t x - x\| = 0\}$$

both have non-empty interior and form a partition of X .

2 Proof intuition

The separable proof uses an Ovsepian–Pelczynski bounded biorthogonal sequence whose linear span is dense in X . Density is used to show that a diagonal operator S , with eigenvalues roots of unity tending rapidly to 1, satisfies $S^{2m_n} x \rightarrow x$ for every $x \in X$.

For a nonseparable space we cannot ask a countable sequence to span densely. However, every infinite-dimensional Banach space contains a normalized basic sequence (e_n) , and its coordinate functionals extend to uniformly bounded functionals on X . If the periods m_n are chosen much faster than in the separable proof, then $S^{2m_n} \rightarrow I$ in operator norm, not merely strongly on a dense span. That operator-norm return is exactly what removes the separability hypothesis. The growth side of Augé's proof is unchanged: a finite-dimensional separating lemma supplies functionals f_n which grow to infinity off a prescribed union of lines and return arbitrarily close to zero on it.

3 Preliminaries

We use two standard facts. First, every infinite-dimensional Banach space contains a normalized basic sequence. Hence there are vectors $(e_n) \subset X$ with $\|e_n\| = 1$ and functionals $e_n^* \in X^*$ such that

$$e_n^*(e_m) = \delta_{nm}, \quad C := \sup_n \|e_n^*\| < \infty.$$

Indeed the coordinate functionals of the basic sequence are uniformly bounded on its closed span, and Hahn–Banach extends them to X .

Second, we use Augé's finite-dimensional separating lemma [1, Proposition 2.1]. In the two-dimensional space $E = \text{span}\{e_1, e_2\}$, let

$$F = \{ae_1 + be_2 : |b| \leq |a|\}.$$

This is a closed proper union of one-dimensional $\mathbb{K}$ -subspaces of E , and both F and $E \setminus F$ have non-empty relative interior. The lemma gives linear forms $f_k \in E^*$, indexed below by $k \geq 3$, such that

$$y \notin F \Rightarrow |f_k(y)| \rightarrow \infty, \quad y \in F \Rightarrow \lim_{k \rightarrow \infty} |f_k(y)| = 0.$$

Choose positive integers m_k , with $m_1 = m_2 = 1$ and $m_k \mid m_{k+1}$, so rapidly increasing that

$$\sum_{k=3}^{\infty} \frac{\|f_k\|}{m_{k-1}} < \infty, \quad \sum_{k=3}^{\infty} \frac{m_{k-2}}{m_{k-1}} \|f_k\| < \infty, \quad \alpha_n := \sum_{k>n} \frac{m_n}{m_k} \rightarrow 0. \quad (1)$$

Such a sequence is obtained recursively, choosing each next multiple large enough to make the displayed tails as small as desired.

4 The complex case

Theorem 1. *Let X be an infinite-dimensional complex Banach space. There is $R \in \mathcal{L}(X)$ such that $R = I + N$ with N nuclear and*

$$A_R = \{x : Px \notin F\}, \quad B_R = \{x : Px \in F\},$$

where $Px = e_1^*(x)e_1 + e_2^*(x)e_2$. In particular A_R and B_R have non-empty interior and form a partition of X .

Proof. For $k \geq 3$ put $\lambda_k = \exp(\pi i/m_k)$, and define

$$Sx = x + \sum_{k=3}^{\infty} (\lambda_k - 1)e_k^*(x)e_k,$$

$$Rx = Sx + \sum_{k=3}^{\infty} \frac{1}{m_{k-1}} f_k(Px) e_k.$$

The two series converge in operator norm by (1) and $|\lambda_k - 1| \leq \pi/m_k$. Moreover

$$R - I = \sum_{k=3}^{\infty} e_k \otimes \left((\lambda_k - 1) e_k^* + \frac{1}{m_{k-1}} f_k \circ P \right),$$

and the sum of the norms of these rank-one summands is finite. Thus $R - I$ is nuclear.

For $t \geq 1$, set

$$\lambda_{k,t} = 1 + \lambda_k + \dots + \lambda_k^{t-1}.$$

Since $Pe_k = 0$ for $k \geq 3$ and $PS^t = P$, induction gives

$$R^t x = S^t x + \sum_{k=3}^{\infty} \frac{\lambda_{k,t}}{m_{k-1}} f_k(Px) e_k. \quad (2)$$

Also

$$S^t x = x + \sum_{k=3}^{\infty} (\lambda_k^t - 1) e_k^*(x) e_k.$$

If $q_n = 2m_n$, then $\lambda_k^{q_n} = 1$ for $k \leq n$, and hence

$$\|S^{q_n} - I\| \leq C \sum_{k>n} |\lambda_k^{q_n} - 1| \leq 2\pi C \sum_{k>n} \frac{m_n}{m_k} = 2\pi C \alpha_n \rightarrow 0.$$

Let $y = Px \notin F$. For $m_{k-1} \leq t \leq m_k$,

$$|\lambda_{k,t}| = \frac{|\sin(\pi t/(2m_k))|}{|\sin(\pi/(2m_k))|} \geq \frac{2}{\pi} m_{k-1}.$$

Applying e_k^* to (2),

$$C\|R^t x\| \geq |e_k^*(R^t x)| \geq \frac{|\lambda_{k,t}|}{m_{k-1}} |f_k(y)| - |e_k^*(x)| \geq \frac{2}{\pi} |f_k(y)| - C\|x\|.$$

As $t \rightarrow \infty$, the corresponding $k \rightarrow \infty$, and $|f_k(y)| \rightarrow \infty$. Thus $\|R^t x\| \rightarrow \infty$, so $x \in A_R$.

Now let $y = Px \in F$. Choose $k_j \rightarrow \infty$ such that $|f_{k_j}(y)| \rightarrow 0$, and put $q_j = 2m_{k_j-1}$. The S -part satisfies $S^{q_j} x \rightarrow x$. For the perturbation part in (2), the terms with $k < k_j$ vanish because $\lambda_{k,q_j} = 0$. The $k = k_j$ term has norm at most $2|f_{k_j}(y)|$. The tail $k > k_j$ is bounded by

$$2\|P\|\|x\| \sum_{k>k_j} \frac{m_{k_j-1}}{m_{k-1}} \|f_k\| \leq 2\|P\|\|x\| \sum_{k>k_j} \frac{m_{k-2}}{m_{k-1}} \|f_k\|,$$

which tends to 0 by (1). Therefore $\lim_t \|R^t x - x\| = 0$, so $x \in B_R$.

We have proved $P^{-1}(E \setminus F) \subset A_R$ and $P^{-1}(F) \subset B_R$. These two preimages partition X , and a vector in B_R cannot have norm tending to infinity along the whole orbit. Hence the displayed equalities hold. Since P is a continuous projection onto the two-dimensional space E , and both F and $E \setminus F$ have non-empty relative interior, both A_R and B_R have non-empty interior in X . $\square$

5 The real case

Theorem 2. *The same conclusion holds for every infinite-dimensional real Banach space.*

Proof. Use the same first two vectors e_1, e_2 , the same projection P , the same set $F \subset \text{span}\{e_1, e_2\}$, and separating forms $(f_k)_{k \geq 2}$. Group the remaining basic vectors into pairs (e_{2k-1}, e_{2k}) , $k \geq 2$, and choose the periods m_k satisfying the analogues of (1).

Let $\theta_k = \pi/m_k$. Define S so that it is the identity outside the chosen coordinates and, on each pair, acts by the rotation

$$e_{2k-1} \mapsto \cos \theta_k e_{2k-1} + \sin \theta_k e_{2k}, \quad e_{2k} \mapsto -\sin \theta_k e_{2k-1} + \cos \theta_k e_{2k}.$$

In operator form this differs from the identity by an absolutely summable rank-one series, since $\sum_k |\sin \theta_k| < \infty$ and $\sum_k |\cos \theta_k - 1| < \infty$. Set

$$Rx = Sx + \sum_{k=2}^{\infty} \frac{1}{m_{k-1}} f_k(Px) e_{2k}.$$

Again $R - I$ is nuclear.

Let

$$\mu_{k,t} = \sum_{\ell=0}^{t-1} S^\ell e_{2k}.$$

The same induction as before gives

$$R^t x = S^t x + \sum_{k=2}^{\infty} \frac{1}{m_{k-1}} f_k(Px) \mu_{k,t}.$$

The rotation period is $2m_k$, so $\mu_{k,2m_n} = 0$ whenever $k \leq n$. The choice of periods also gives $S^{2m_n} \rightarrow I$ in operator norm: for $k > n$, the norm of the difference between the $2m_n$ -th rotation and the identity is $O(m_n/m_k)$, and the tail is controlled by $\alpha_n \rightarrow 0$.

For $m_{k-1} \leq t \leq m_k$, the two coordinates of $\mu_{k,t}$ are the real and imaginary parts, up to signs, of $\sum_{\ell=0}^{t-1} e^{i\ell\theta_k}$. Hence at least one coordinate has absolute value at least

$$\frac{1}{\sqrt{2}} \frac{|\sin(\pi t/(2m_k))|}{|\sin(\pi/(2m_k))|} \geq \frac{2}{\pi\sqrt{2}} m_{k-1}.$$

Applying the corresponding coordinate functional e_{2k-1}^* or e_{2k}^* yields

$$C\|R^t x\| \geq c |f_k(Px)| - C\|x\|$$

with an absolute constant $c > 0$. This proves divergence for $Px \notin F$.

If $Px \in F$, choose k_j with $f_{k_j}(Px) \rightarrow 0$ and take $q_j = 2m_{k_j-1}$. The blocks with $k < k_j$ vanish at time q_j , the k_j -block is bounded by a constant times $|f_{k_j}(Px)|$, and the tail is bounded by

$$C'\|P\|\|x\| \sum_{k > k_j} \frac{m_{k-2}}{m_{k-1}} \|f_k\|,$$

which tends to zero. Thus $R^{q_j} x \rightarrow x$. The same partition and interior argument as in the complex case completes the proof. $\square$

6 Verification and novelty check

The proof uses no numerical computation. The packet includes `code/crop_open_problem.py`, which only renders the page-8 source crop from `source_paper.pdf`.

Local duplicate checks in the run indexes found no previous packet for arXiv:1204.2044 or for this arbitrary-Banach-space strengthening. External web searches on 2026-06-20 used the exact title, arXiv id, Remark 3.8 wording, “operators with the properties stated in Theorem 1.1 arbitrary Banach spaces”, “Prajitura’s conjecture arbitrary Banach spaces”, and nearby phrases involving A_R , $I + K$, and nuclear perturbations. These searches found the source paper and the earlier Hajek–Smith symmetric-basis paper, but no later exact answer to Remark 3.8. This is a bounded novelty check, not a claim of exhaustive bibliographic certainty.

7 Human review recommendation

Review the operator-norm return estimate $S^{2m_n} \rightarrow I$, since that is the new point replacing separability. Also check that the real rotation-block translation preserves the lower-coordinate estimate and the nuclearity of $R - I$. If those estimates are accepted, the packet gives a full positive answer to Augé’s arbitrary Banach space conjecture in Remark 3.8.

References

- [1] J.-M. Augé, *Linear operators with wild dynamics*, arXiv:1204.2044, 2012.
- [2] P. Hájek and R. J. Smith, *Operator machines on directed graphs*, Integral Equations Operator Theory 67 (2010), 15–31.
- [3] I. Singer, *Bases in Banach Spaces I*, Springer, 1970.